



Proposal & Intake Form

Bring Your Own Capstone

How your capstone is approved — complete every section

This form will be used to assess your proposal for acceptance or rejection — complete every section. Vague or incomplete proposals are returned for revision, and once accepted your problem and dataset are locked (scope can be scaled down later, but no switching domain or dataset).

1 · Learner & project

Learner name	Brijesh Arora
Email ID	brijesharora01@gmail.com
Program and track	Master's In Data Science (upGrad and IIITB Program) Track: DA
Proposed project title	Diagnosing Operational Drivers of Customer Repeat Purchasing in E-Commerce: A Data-Driven Retention and Revenue Opportunity Analysis.

2 · Problem & business value

Criterion 1 — a measurable organisational problem, not theoretical curiosity.

Business problem / pain point	The e-commerce marketplace has a very low repeat-purchase rate, but the business lacks clear evidence on whether operational factors such as late delivery, shipping charges, and customer satisfaction materially influence a customer's decision to purchase again. Without quantifying these relationships and their potential revenue impact, management cannot confidently prioritize operational improvements or determine where intervention could generate the greatest retention and revenue benefit.
Why it matters strategically?	Improving repeat purchasing is critical to sustainable e-commerce growth because acquiring new customers is generally more costly than retaining existing ones. Identifying which aspects of the first-order experience are associated with repeat purchasing will help management prioritize operational improvements, quantify potential revenue opportunities, and allocate resources based on evidence rather than assumptions.
Earlier solution attempts and pitfalls	Initial approaches focused on generic customer segmentation, cohort analysis and repeat-purchase prediction. However, these approaches risked temporal leakage/reverse causality by using information from orders placed after the customer's first purchase. The initial use of freight-to-order-value ratio also created strong correlation with order value, making interpretation unreliable. The project was therefore redesigned to use first-order information only , separate freight value from order value, limit formal hypotheses to two, and use Logistic Regression as a confirmatory rather than primary model.

Success metric	Primary: Recoverable Revenue-at-Risk — the estimated portion of GMV associated with non-repeat customers that could realistically be recovered through a feasible, costed intervention, evaluated separately for the delivery and shipping-cost levers. Secondary: Repeat-purchase rate, retention gap between on-time and late first-order cohorts, effect sizes and 95% confidence intervals for the two pre-registered hypotheses, and Segment Expected Repeat Revenue to prioritise higher-value customer segments.
Primary stakeholders	Head of Customer Experience — uses the analysis to prioritise retention interventions and assess the potential revenue impact. Marketplace Operations — evaluates shipping-cost and seller-related operational opportunities. Logistics / Fulfilment — uses delivery-performance findings to identify areas where service improvements may support retention. Marketing / Customer Retention — uses the findings to assess whether retention-focused investment should complement customer-acquisition spending.

3 · Data — availability & legal provenance

Criterion 2 — you must already have the data. “To be sourced later” is rejected.

Dataset name & source	Brazilian E-Commerce Public Dataset by Olist — sourced from Kaggle , originally provided by Olist. The dataset contains approximately 100,000 real, anonymised e-commerce orders from 2016–2018 across nine relational tables covering orders, customers, items, payments, reviews, products, sellers, geolocation and product-category translation.
Link / how it is attached	Source: Kaggle — Brazilian E-Commerce Public Dataset by Olist How obtained: Downloaded the complete dataset and stored locally; all nine CSV files are already in hand and have been validated for the project. The dataset contains approximately 100,000 orders across multiple relational tables. Attachment: The downloaded ZIP/CSV files will be retained as the project's raw data and are not to be sourced later . This satisfies the BYOC requirement that data must already be available before proposal submission.
How you obtained it	Downloaded directly from the Kaggle dataset page to my local machine.
Licence & legal status	Licence: CC BY-NC-SA 4.0 as listed on the current Kaggle dataset page. Public, anonymised dataset provided by Olist and available through Kaggle. The dataset is described by Olist as real commercial e-commerce data with identifying information anonymised.
PII / IP check	No direct PII identified. The dataset is publicly available and customer identifiers are anonymised/encoded rather than containing names,

	email addresses, phone numbers or physical addresses.
Data format and size	Format: 9 relational CSV files covering orders, order items, customers, reviews, payments, products, sellers, geolocation and product-category translation. Size: 99,441 orders, 112,650 order-item records, 99,441 customers, 99,224 reviews, 103,886 payment records, 32,951 products, 3,095 sellers, 1,000,163 geolocation records and 71 category translations. Scale: Approximately 1 million+ rows across the nine tables, providing sufficient relational complexity for DuckDB-based ingestion, joining and analytical processing.

☒ I confirm I already have access to this data now — it is not “to be sourced later.”

4 · Scope & solution approach

Criterion 4 — must meet your track Scope Specification benchmarks.

Scope-Spec stages & components you will cover	Hypothesis Generation — Define the measurable business problem, KPIs, research questions and two statistically testable hypotheses before analysis. Ingestion & Profiling — Ingest the nine Olist CSV tables, profile distributions and missingness, validate data quality, and document treatment decisions. High-Performance Processing — Use DuckDB to clean, join and aggregate the relational data and create a reproducible first-order customer analytical dataset. Diagnostic Modelling — Apply descriptive analysis, cohort/value segmentation, Chi-square, Mann-Whitney U, effect sizes, odds ratios and 95% confidence intervals to evaluate the hypotheses. Confirmatory Modelling — Use a single interpretable Logistic Regression to assess whether the identified first-order factors carry predictive signal, with appropriate treatment of class imbalance and multicollinearity. Business Impact Analysis — Estimate Recoverable Revenue-at-Risk and Segment Expected Repeat Revenue under clearly documented assumptions, without pre-committing to a positive business outcome. Interactive Visualisation — Build a filterable Power BI/Tableau dashboard for stakeholders to explore retention, operational drivers, segments and potential business impact. Reproducibility & Communication — Maintain a Git repository with modular SQL/Python code, pinned dependencies and README; deliver the required technical report and executive presentation.
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Tools / stack (open-source, local)

NOTE: upGrad is not liable to provide access or support for tools or technologies

Python (data preparation, statistical analysis and modelling), **DuckDB** (high-performance SQL-based ingestion, joining and aggregation), **Jupyter Notebook** (analysis and documentation), **Power BI / Tableau** (interactive stakeholder dashboard), and **Git/GitHub** (version control and reproducibility).

No cloud infrastructure or proprietary employer systems will be used.

5 · Week-by-week plan

Criterion 3 — a realistic plan fitting ~30–40 hours over four weeks.

Week	Milestone / deliverable	Est. hours
Week 1	Ingest and profile all nine Olist tables in DuckDB; validate joins and keys; document missingness/outliers; build the first-order customer analytical dataset; validate repeat-purchase definition and target. Deliverable: Clean, validated analytical dataset.	8 – 10 hrs.
Week 2	Perform descriptive analysis and value segmentation; execute H1 and H2 tests; calculate effect sizes, odds ratios and 95% CIs; conduct correlation/multicollinearity checks; fit the single confirmatory Logistic Regression. Deliverable: Validated statistical findings and confirmatory model.	9 – 10 hrs.
Week 3	Calculate Recoverable Revenue-at-Risk and Segment Expected Repeat Revenue using documented assumptions; develop the analytical dashboard; translate validated findings into candidate business interventions. Deliverable: Business-impact analysis and working interactive dashboard.	8 – 10 hrs.
Week 4	Complete dashboard; perform QA and sensitivity checks; finalise recommendations and limitations; write technical report; prepare executive presentation; clean GitHub repository and README. Deliverable: Submission-ready report, executive deck and reproducible repository.	6 – 8 hrs.

6 • Deliverables

You will submit all three artifacts on the upGrad platform: ☒ Technical report (10–15 pages) ☒
Executive presentation (8–12 slides) ☒ ZIP / GitHub repository + README

7 • Feasibility self-check

- ☒ The scope realistically fits ~30–40 hours over four weeks.
- ☒ My data is in hand now and legally usable.
- ☒ My project meets the “minimum to be accepted” in my Scope Specification.
- ☒ I have a primary measurable success metric with supporting secondary metrics.
- ☒ This is an original problem — not a cloned tutorial or a toy dataset.

8 • Declaration

- ☒ This is my original work. ☒ My data is legally usable. ☒ Scope fits ~30–40 hours over four weeks.

Signature: 

Date: 18-08-2026